STRATEGIC PARTNERSHIP OFFER

Open Bioengineering & AI Safety Architectures

“Althea” (CRISPR-TTP) + “AMAGI” (Hardware-Enforced AI Safety)

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Legal Entities: LLC “Althea” (Russia) — active · AGIAM Technologies Ltd (UK) — in formation (inactive shell)

> **Terminology note (binding for this document).** The acronym TTP is used historically in two different ecosystems. Per the AMAGI Master Glossary v1.0 §8.3 (DOI: 10.5281/zenodo.23797780), in the AMAGI corpus **TTP canonically denotes the Tautological Truth Predicate**, and the safety structure formerly expanded as “Trigger → Programmable Delay → Localized Activation” is named the **Temporal Triad** (resolution D-09). In the Althea bioengineering corpus, **TTP denotes Temporally Programmed Delivery**. The two terms are distinct and are not conflated anywhere in this document (see §4).

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1. Executive Summary

We present two complementary, fully open architectures that together form a complete ecosystem for trustworthy, life-critical autonomous systems. As of this version, each track carries its own class of independent evidence: **Althea — peer review in Frontiers in Genetics (Scopus CiteScore Q1); Amagi — exhaustive formal verification**, both public and DOI-stamped.

1.1 “Althea” — CRISPR-TTP Bioengineering Platform

A fully open (CC0) bioengineering architecture for the treatment of fusion-driven cancers, with an initial focus on Ewing Sarcoma — the second most common bone cancer in children and young adults.

Peer-reviewed publication (independent validation of the architecture):

> Burlai AM (2026). Genomic innovation in precision oncology: integrated CRISPR-TTP bioengineering architecture for Ewing Sarcoma (version 4.0 — complete architectural specification). **Frontiers in Genetics**, section Cancer Genetics and Oncogenomics; article type: **Brief Research Report**. DOI: 10.3389/fgene.2026.1727708 · PMID: 41743405 · PMCID: PMC12931889. Received 18 Oct 2025 · Revised 01 Dec 2025 · Accepted 20 Jan 2026 · Published 11 Feb 2026. Reviewed by **Biaoru Li (Augusta University, USA)** and **Peter Anderson (Cleveland**

> **Clinic, USA)**; edited by **Mallikarjuna Rao Gedda (National Heart, Lung, and Blood Institute, NIH, USA)**. Open Access, CC BY.

All efficacy figures in the article are **computational projections** ("Projected Efficacy"; "in silico modeling predicts") — they are not experimental results. See the evidence-class table (§5) and the risk register (§10, R1).

- Core technologies:
  - CRISPR-Cas9 genome editing of autologous dendritic cells (>94% efficiency; 17 strong MHC-I + 9 strong MHC-II binders, NetMHCpan 4.1);
  - Temporally Programmed Delivery (TTP) via MR-guided Focused Ultrasound (FUS) activating HOF nanoparticles (27 ± 3 nm; 1–2 mm spatial resolution);
  - AI Co-pilot for real-time personalization, off-target risk mitigation, and closed-loop therapeutic control (89.3% sgRNA prediction accuracy).
- Core publications: the peer-reviewed article (above); full architectural blueprint v4.0 (DOI: 10.5281/zenodo.17780623 — the article's data statement points here); clinical SOP v2.0 (DOI: 10.5281/zenodo.17676423); TTP v3.0 delivery architecture (DOI: 10.5281/zenodo.17237907); CRISPR Autovaccination v3.0 (DOI: 10.5281/zenodo.17246573). Full list in §8.1.

1.2 "Amagi" — Hardware-Enforced AI Safety Architecture

A reference architecture designed to provide structural safety assurances for high-risk AI systems, transforming regulatory compliance from a procedural burden into architectural guarantees.

- Core principles: six immutable invariants (Physical Non-Bypassability, Irreducible TCB, Separation of Control, Immutability of Policy, Complete Mediation, Complete Audit Integrity);
- **Temporal Triad** (Trigger → Programmable Delay → Localized Activation) as a universal safety primitive (naming per Master Glossary v1.0 §8.3, resolution D-09 — see the Terminology note);
- Hardware components: SCC (Security Council Core), NIPU (Neuro-Immune Processing Unit), MEM (Self-Protecting Memory), Amagi-Q (post-quantum crypto);
- **Formally verified:** the SCC Promela model (scc_amagi_v3) has been exhaustively model-checked with SPIN 6.5.2 — **20 verification runs covering 15 unique claims, all verified** (safety, liveness, and counterexample suites; state space 25,927,680 states; parameters W=24, M=2,000,000). The complete public dataset — models, logs, trails, checksums — is published as DOI: 10.5281/zenodo.23809098 (v3.1, October 30, 2026) and is independently re-runnable;
- Regulatory alignment: EU AI Act; NIST AI RMF 1.0 (Profile v1.1); IEC 61508 / ISO 26262 (ASIL-D) assurance artifacts (HARA v1.1, FMEDA v1.1, Traceability Matrix v1.1);
- Core publications (updated to the October 2026 corpus): full list in §8.2.

2. Current Status & Achievements (October 2026)

2.1 "Althea" (CRISPR-TTP)

Area	Status	Reference
Peer-reviewed publication	Frontiers in Genetics — CiteScore 6.9 (Q1, Scopus), JIF 3.0 (Q2, WoS), per the journal's metrics page (Oct 2026); Brief Research Report, published Feb 11, 2026	DOI: 10.3389/fgene.2026.1727708; PMID 41743405; PMCID PMC12931889
Full architectural blueprint	v4.0, open (CC0), the article's cited data source	10.5281/zenodo.17780623
Clinical protocol	21-day GMP-compatible SOP v2.0	10.5281/zenodo.17676423

Area	Status	Reference
Regulatory dialogue	EMA: ATMP classification determined (Feb 11, 2026); FDA CBER: INTERACT meeting submitted (Feb 18, 2026); Russian MoH: expert evaluation completed (Feb 19, 2026) — status as reported in SPO v1.0 and carried forward	—
Foundational architectures	TTP v3.0 (delivery); CRISPR Autovaccination v3.0	10.5281/zenodo.17237907; 10.5281/zenodo.17246573
Next mandatory step	Wet-lab (in vitro / in vivo) validation of the in silico projections — this is the explicit subject of ask **A1** (§6)	—

2.2 “Amagi” (AI Safety)

Area	Status	Reference
Architecture specification	Amagi Framework v2.0 (June 2026)	10.5281/zenodo.20753400
Open standard	Amagi Open Specification v1.1	10.5281/zenodo.20753403
Component specifications	NIPU v1.1; MEM v1.1; Architectural Class Definition (IAA) v1.1	10.5281/zenodo.20753401; 10.5281/zenodo.20753402; 10.5281/zenodo.20145198
Regulatory alignment	NIST AI RMF Profile v1.1	10.5281/zenodo.20753404
Assurance & certification	HARA v1.1; FMEDA v1.1; Traceability Matrix v1.1; Master Parameter Sheet v2.1 (all Oct 30, 2026)	10.5281/zenodo.23797808; 10.5281/zenodo.23797807; 10.5281/zenodo.23797811; 10.5281/zenodo.23797802
Terminology governance	Master Glossary v1.0 — canonical terminology register, resolutions D-01…D-16 (Oct 30, 2026)	10.5281/zenodo.23797780
Version control	Master Version Control Document v1.1 — certification baseline (June 2026 snapshot) (Oct 30, 2026)	10.5281/zenodo.23797812
Formal verification	SCC verification package v3.1: 20 SPIN runs / 15 unique claims, all verified (Oct 30, 2026)	10.5281/zenodo.23809098
Legal & licensing	AGIAM Charter v1.2; S-APL v2.0; Trade Secrets Notice v1.0	10.5281/zenodo.20753872; 10.5281/zenodo.20753406; 10.5281/zenodo.17886699
Journal track	TCAD manuscript rejected 2026-09-22; a revised submission with a response letter is in preparation (Master Version Control Document v1.1 §6.4)	—

3. The Unmet Need — Why Both Architectures Are Necessary

3.1 Medical Challenge

Ewing Sarcoma is a devastating paediatric malignancy. Despite multimodal therapy, 5-year survival for metastatic disease remains below 30%. The pathognomonic EWSR1-FLI1 fusion oncogene is an ideal tumour-specific target, yet it has proven undruggable by conventional pharmacology. The tumour microenvironment is immunologically “cold”, with downregulated HLA-I and high Treg infiltration.

3.2 AI Safety Challenge

Modern AI systems controlling life-critical functions are inherently unpredictable. Traditional software-based safety measures can be bypassed, creating unacceptable risks. Regulators (EU AI Act, NIST, FDA) now demand structural guarantees, not just procedural promises.

Althea addresses the biological complexity. Amagi addresses the computational risk. Together, they enable a future where autonomous therapeutic systems are both effective and designed for trustworthiness.

4. One Architectural Pattern, Two Instantiations — Why Together

Both architectures gate a consequential action behind an explicit, verifiable, temporally structured control step:

- **Althea** gates the release of a therapeutic payload. Temporally Programmed Delivery activates HOF nanoparticles by MR-guided FUS at a programmed time and location: the genotoxic consequence does not occur until its preconditions — targeting, dosing, timing — are verified.
- **Amagi** gates the release of an AI actuation. The Temporal Triad (Trigger → Programmable Delay → Localized Activation) is enforced in hardware: the actuated consequence cannot occur until the policy chain has explicitly and irreversibly authorized it.

The shared pattern — **gating the release of a consequential action behind multi-stage, verifiable control** — is one architectural philosophy applied to two of the highest-stakes domains: the genome and machine actuation.

The shared acronym TTP is historical, not taxonomic. Per the Master Glossary v1.0 §8.3, in the AMAGI corpus TTP canonically denotes the Tautological Truth Predicate and the triad is the Temporal Triad; in the Althea corpus TTP denotes Temporally Programmed Delivery. This document disambiguates the terms explicitly wherever they co-occur, as the Glossary requires of cross-ecosystem documents.

5. What We Have Already Achieved — With Evidence Classes

Every performance figure below carries an explicit evidence class. Computational and synthesis-stage figures are **not** experimental or field measurements.

Evidence class	Althea	Amagi
Peer-reviewed architecture	Frontiers in Genetics (2026), Brief Research Report — architecture reviewed and accepted	— (journal submission rejected 2026-09-22; revised submission in preparation)
Formal verification (exhaustive model checking)	—	SCC v3.1: 20 SPIN runs, 15 unique claims, all verified; state space 25,927,680 states
Computational predictions & analysis figures	Tumor growth inhibition ~96.3%; median survival improvement ~65%; CD8+ T-cell infiltration ×3.2; CRISPR editing >94%; sgRNA prediction 89.3% — all in silico	Response latencies 12–35 µs (TCB-only envelope); SFF 93% (system-level) — per Master Version Control Document v1.1 §6.3, fixed as immutable with their classes
Hardware evidence	—	Proof-of-logic: post-synthesis FPGA evidence, single Xilinx XC7A35T device (a board-level measurement campaign has **not** been performed — designated Phase 2)
Open specifications	v4.0 blueprint; SOP v2.0; TTP v3.0; Autovaccination v3.0 (CC0)	Framework v2.0; Open Spec v1.1; NIPU/MEM v1.1; NIST Profile v1.1; HARA/FMEDA/TM v1.1; MPS v2.1 (CC BY-NC 4.0 / S-APL)
Planned (not yet done)	Wet-lab and in vivo validation; Phase I/II trials	Board-level measurement campaign; ASIC implementation; standard-body adoption

6. What We Are Looking For — Concrete Asks

We invite partners to join us in translating both architectural blueprints into experimental and clinical reality. The menu of v1.0 is upgraded to concrete asks; horizons are proposals to be finalized with each partner.

A1 — Althea (highest priority): wet-lab validation partner. A CRO or academic lab for in vitro / in vivo validation of the projected efficacy figures of §5. Horizon: first wet-lab milestone within 6–12 months of agreement. Deliverable: a validated — or falsified, equally publishable — preclinical dataset, co-authored. The partner gets a CC0 head start on a peer-reviewed architecture, co-authorship, and a grant-ready evidence base.

A2 — Althea: GMP manufacturing partner. Centres for dendritic-cell production and CRISPR editing, to prepare the 21-day GMP-compatible SOP v2.0 for first-in-human conditions.

A3 — Amagi (highest priority): hardware implementation partner. A semiconductor/FPGA company for SCC IP-core implementation and a board-level evaluation campaign on the path to ASIC. Current evidence: post-synthesis proof-of-logic on a single Xilinx Artix-7 class device (XC7A35T). Horizon: board-level campaign within 6–12 months. Deliverable: measured board-level evidence closing the Phase 2 gap defined in the Master Version Control Document v1.1 §6.3.

A4 — Amagi: regulatory & standards pilot. Certification bodies and consultants for pilot mappings of the open specifications onto IEC 61508 / ISO 26262 / EU AI Act conformity workflows.

A5 — Both: grant consortia. Co-application for Horizon Europe, EIC, and comparable programmes, with Althea and Amagi as **separate work packages with separate evidence classes** (§5).

A6 — Both: advisory board. Members with clinical, regulatory (ATMP / AI), or semiconductor backgrounds.

6.1 Flexible Partnership Formats

- Scientific collaboration (joint publications);
- Service agreements (CRO, manufacturing);
- Grant consortia;
- Technology / IP-core licensing (S-APL v2.0);
- Equity-free sponsorship (philanthropic or corporate);
- Advisory board membership.

7. Why Partner With Us?

- **Peer-reviewed track record (Althea):** the core architecture passed independent review in Frontiers in Genetics (Scopus CiteScore Q1) — examined by reviewers from Augusta University and Cleveland Clinic under an NHLBI/NIH editor. The figures were inspected, not self-asserted.
- **Formally verified track record (Amagi):** every safety claim is machine-checkable and re-runnable from the public dataset (SPIN 6.5.2; W=24, M=2,000,000; 20 runs / 15 claims).
- **Open by design:** no IP battles — all knowledge is freely available as DOI-stamped prior art.
- **Regulator-ready (Althea):** already engaged with EMA (Feb 11), FDA (Feb 18), and Russian MoH (Feb 19) — as reported in SPO v1.0, carried forward.
- **Standards-aligned (Amagi):** a complete NIST AI RMF Profile v1.1 and IEC 61508 / ISO 26262 assurance artifacts.
- **Honest evidence discipline:** figures carry explicit evidence classes (§5); this offer includes its own risk register (§10).
- **Modular & adaptable:** Althea can be repurposed for other malignancies; Amagi applies to any high-risk AI.

- **Ethically grounded:** dedicated to the memory of Althea (Irina Neboga) — a humanitarian mission.
- **Complete ecosystem:** from therapeutic mechanism to hardware-enforced AI safety — one gated-consequence pattern in two domains (§4).

8. Complete List of Published Works

8.1 Althea / CRISPR-TTP Series

1. **Official publication (v4.0):** Burlai AM (2026), Frontiers in Genetics — 10.3389/fgene.2026.1727708 (PMID 41743405; PMCID PMC12931889; published Feb 11, 2026)
2. Official publication record on Zenodo (article PDF) — 10.5281/zenodo.18605188
3. Architectural blueprint v4.0 (preprint; the article's data source) — 10.5281/zenodo.17780623
4. SOP "Althea" v2.0 (21-day GMP-compatible clinical protocol) — 10.5281/zenodo.17676423
5. Complete Clinical Implementation Protocol v1.0 — 10.5281/zenodo.17537570
6. Genomic Innovation v3.0 (manuscript) — 10.5281/zenodo.17526578
7. An Integrated Bioengineering Architecture for Ewing Sarcoma (Oct 2025) — 10.5281/zenodo.17308401
8. URGENT MEDICAL CASE: CRISPR Autovaccination for Metastatic Ewing Sarcoma — 10.5281/zenodo.17247084
9. CRISPR-Induced Autovaccination v3.0: Universal Therapeutic Architecture Declaration — 10.5281/zenodo.17246573
10. Triggered Therapy Platform (TTP): Universal Architecture Declaration v3.0 — 10.5281/zenodo.17237907

8.2 Amagi / AGIAM Series (updated October 2026)

1. Amagi Framework v2.0 — 10.5281/zenodo.20753400
2. Amagi Open Specification v1.1 — 10.5281/zenodo.20753403
3. Amagi-NIPU v1.1 — 10.5281/zenodo.20753401
4. Amagi-MEM v1.1 — 10.5281/zenodo.20753402
5. Amagi Architectural Class Definition (IAA) v1.1 — 10.5281/zenodo.20145198
6. NIST AI RMF Profile v1.1 — 10.5281/zenodo.20753404
7. Amagi HARA v1.1 — 10.5281/zenodo.23797808
8. Amagi FMEDA v1.1 — 10.5281/zenodo.23797807
9. Amagi Traceability Matrix v1.1 — 10.5281/zenodo.23797811
10. AMAGI Master Parameter Sheet v2.1 — 10.5281/zenodo.23797802
11. AMAGI Master Glossary v1.0 — 10.5281/zenodo.23797780
12. AMAGI Master Version Control Document v1.1 — 10.5281/zenodo.23797812
13. AMAGI Supplementary SCC Verification Package v3.1 — 10.5281/zenodo.23809098
14. AGIAM Charter v1.2 — 10.5281/zenodo.20753872
15. Amagi S-APL v2.0 — 10.5281/zenodo.20753406
16. Amagi Framework Trade Secrets Notice v1.0 — 10.5281/zenodo.17886699

9. Next Steps

1. Initial contact: send an expression of interest to lexintrud@gmail.com (subject: "Partnership Inquiry — Althea/Amagi", or "Althea" / "Amagi" for a single track).
2. NDA & information exchange (if required).
3. Discussion of a specific collaboration format and ask (A1–A6).
4. Formal agreement (partnership, service, or licensing contract).

10. Risk Register

Published in this offer so that partners find the risks — and the answers to them — here first.

#	Risk	Track	Mitigation / status
R1	All efficacy figures are in silico projections; no wet-lab validation has been performed. Peer review validated the coherence of the architecture, not its biology.	Althea	Ask A1 exists precisely to close this gap; the article marks every figure as “Projected Efficacy” / “in silico modeling predicts”
R2	The journal article is a **Brief Research Report** (short format).	Althea	The full v4.0 architectural blueprint is openly available and is the article’s cited data source (10.5281/zenodo.17780623)
R3	The FUS + HOF delivery platform is unproven in vivo.	Althea	Open TTP v3.0 architecture; asks A1/A2; SOP v2.0 ready for GMP conditions
R4	Long ATMP regulatory timelines; regulatory status was last publicly reported in February 2026.	Althea	EMA ATMP classification determined; FDA INTERACT submitted; clinical SOP v2.0 in place
R5	No journal peer-review yet for the AMAGI architecture; the TCAD manuscript was rejected on 2026-09-22; a revised submission is in preparation.	Amagi	Formal verification is public and independently re-runnable (10.5281/zenodo.23809098); the Master Version Control Document v1.1 §6.4 makes the Canonical-Reference-Implementation status independent of the review outcome
R6	Hardware evidence is proof-of-logic only: post-synthesis FPGA evidence on a single XC7A35T; no board-level measurement campaign (designated Phase 2).	Amagi	Ask A3 exists precisely to close this gap
R7	Adoption risk: the value of the AMAGI architecture depends on standard-body and vendor adoption.	Amagi	NIST AI RMF Profile v1.1; IEC 61508 / ISO 26262 mappings; open specifications lower the adoption cost
R8	Single-author dependency; the AMAGI-side legal entity is in formation (not an active legal entity), so contracting requires planning.	Both	The open corpus guarantees continuity independent of any single entity or company; AGIAM Charter v1.2; sustainment and succession documentation in preparation; LLC “Althea” (RF) is active
R9	Two domains in one offer may dilute focus perception for a specialist audience.	Portfolio	Tracks and evidence classes are separated throughout (§1, §2, §5); single-track offers (Althea only / Amagi only) are available on request

11. Contact & Legal Information

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Legal Entities:

- LLC “Althea” (Russia): Active (INN 4217214140, OGRN 1264200000900)
- AGIAM Technologies Ltd (UK): in formation (inactive shell, not an active legal entity)

Licensing:

- Althea (medical): CC0 1.0 Universal — freely available for any purpose.
- Amagi (AI safety): CC BY-NC 4.0; S-APL v2.0 required for commercial use.
- Hard IP Suite: protected as trade secrets; access requires NDA + S-APL.

Disclaimer

This document describes architectural concepts and research frameworks. It does not constitute medical advice, regulatory approval, or investment solicitation. All clinical and hardware implementations require independent validation and regulatory authorization.

Funding Statement

No external funding was received for the preparation of this document.

Conflict of Interest

The author is the founder of LLC “Althea” and of AGIAM Technologies Ltd (in formation).

This document is released under CC0 1.0 Universal (public domain). Let’s build the future of open-source medicine and trustworthy AI — together.

Version History

- **v1.0 — February 21, 2026** (Zenodo rec 18718186): first public offer.
- **v1.1 — October 31, 2026** (this version): complete bibliographic record of the Frontiers in Genetics publication (DOI, PMID/PMCID, dates, reviewers, editor, article type); AMAGI status updated to the October 2026 corpus (Framework v2.0, Open Spec v1.1, Wave of Oct 30, 2026 — Glossary/MPS/FMEDA/HARA/TM/VCD, SCC formal-verification package v3.1); TTP terminology aligned with Master Glossary v1.0 §8.3 (D-09): “Temporal Triad” naming plus a binding terminology note; new §4 (one architectural pattern, two instantiations); new §5 (evidence-class table); §6 menu upgraded to concrete asks A1–A6 (horizons are proposals); new §10 risk register (R1–R9); TCAD journal-track status disclosed; legal-entity wording updated; publication lists verified record-by-record and corrected (v1.0 items 6–7 duplicated one record family; titles aligned to the actual Zenodo records); journal metrics stated with source and basis — CiteScore 6.9 (Q1, Scopus) and JIF 3.0 (Q2, WoS) per the journal’s metrics page, October 2026 (the v1.0 figure “Q1, CiteScore 6.2” cited the superseded pairing without a source).